SemEval-2026 Task 10: A General Framework based on Pre-trained Language Models for Conspiracy Detection

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Abstract

This paper proposes a general framework based on BERT-based pre-trained models for SemEval-2026 Task 10. Through a systematic evaluation of four BERT variants, we find that the lightweight DistilBERT achieves optimal performance in conspiracy detection, attaining a weighted F1-score of 0.80. Regarding marker extraction, although RoBERTa-Large demonstrates relative superiority, existing architectures struggle to accurately capture the boundaries of high-level semantics, particularly when handling abstract semantic concepts.

1 Introduction

The proliferation of internet platforms has revolutionized how information spreads. Among these, Reddit stands as one of the world's most active communities. With 116 million daily active users (Reddit, Inc., 2025) and its unique "subreddit" architecture, it has become a vital arena for public discourse. However, this interest-based division, combined with users' homophily preferences, easily leads to closed "echo chamber" effects. This, in turn, provides a hotbed for conspiracy theories to thrive and spread rapidly (Papacharissi, 2016). Banas and Miller (Banas and Miller, 2013) define conspiracy theories as causal narratives that attribute events to covert plans orchestrated by a secret cabal. Such narratives not only distort facts but also pose substantial threats to public health and democratic institutions (Diab et al., 2024). Consequently, accurately identifying conspiracy theories within vast volumes of daily discourse and disrupting their propagation is critical.

However, automated detection of conspiracy theories in dynamic and diverse discussion spaces like Reddit faces challenges due to semantic complexity. Introne et al. (Introne et al., 2020) note that conspiracy theories are not mere rumors but intricate narrative structures comprising six key elements: events, actors, goals, actions, consequences, and targets. These implicit and complex semantic characteristics make it difficult for traditional automated methods (e.g., proxy-based methods (Bessi et al., 2015), keyword matching (Kim and Kim, 2023), and pattern matching (Kou et al., 2017)) to capture the contextual logic. The development of deep learning methods, particularly the application of Pre-trained Language Models (PLMs), has significantly enhanced the generalized understanding of textual content (Patwardhan et al., 2023). While past research has achieved notable results in applying pre-trained models to identify various conspiracy theory texts, deconstructing their internal formation mechanisms remains crucial for the interpretation of conspiracy theories. In light of this, SemEval-2026 Task 10: Psycholinguistic Conspiracy Marker Extraction and Detection aims to integrate psychological theory with NLP techniques[1]. By extracting conspiracy markers (Subtask 1) and detecting conspiracy comments (Subtask 2), the task seeks to reveal the expressive mechanisms of conspiracy theories within everyday discourse.

To address these challenges, this study proposes a general conspiracy theory monitoring framework for Reddit comments. By applying BERT-based pre-trained models, it deconstructs the deep semantic features and narrative logic within comment texts. This approach not only achieves precise binary classification of conspiracy comments but also focuses on automatically extracting key psychological markers. Furthermore, by comparing the performance of different models in conspiracy theory classification and marker extraction, we further explore the models' generalization capabilities and interpretability in complex contexts.

[1] https://hide-ous.github.io/semeval2026-psycomark/

2 Related Work

Effective monitoring of conspiracy theories requires a clear understanding of what constitutes a conspiracy narrative and the textual features that distinguish it from mainstream discourse. Scholars define conspiracy theories as causal narratives that attempt to explain significant social or political events as the secret result of powerful actors acting for specific purposes, rather than as random or natural occurrences (Douglas and Sutton, 2018; Banas and Miller, 2013). This distinctive causal logic results in conspiracy theories being highly certainty-oriented, prioritizing closed-ended causal explanations to seek cognitive closure rather than open-ended factual inquiry (Fong et al., 2021). Introne et al. (Introne et al., 2020) specify this as: "A conspiracy theory is a narrative explaining an [event or series of events] that involve [deceptive, coordinated actors] working together to achieve [a goal] through [an action or series of actions] that have consequences that intentionally disenfranchise or harm an [individual or population]."

Numerous strategies has been explored in prior research to identify conspiracy theory. Proxy-based methods utilize proxy variables such as sources, user behavior, and propagation paths to identify conspiracy theory content. Bessi et al. (Bessi et al., 2015) treated data sources as proxy indicators for content nature, with all content on Facebook conspiracy theory pages automatically categorized accordingly. Starbird et al. (Starbird et al., 2019) analyzed user-cited media sources, URL, and information pathways to use "source" as a proxy variable for identifying conspiracy theory. Although this method has low data acquisition costs, it leads to a high number of misclassifications. Keyword-matching methods directly identify content by searching for specific conspiracy theory terms. Kim et al. (Kim and Kim, 2023) uesed keywords to extract QAnon-related conspiracy theories on Facebook. This predefined keyword approach risks overlooking false negatives and often limited to content of specific topics. Regarding pattern-matching-based methods, Kou et al. (Kou et al., 2017) utilized qualitative discourse analysis or syntactic patterns to understand conspiracy theory generation mechanisms, and Samory et al. (Samory and Mitra, 2018) matched text messages against predefined syntactic elements (agent-action-target triplets). This approach has limitations in large-scale automatic detection and cannot be fully automated or generalized to all semantic contexts.

Related research applies deep learning methods to break through the limitations of the aforementioned methods in generalization ability and deep semantic understanding, committing to automatically learning high-dimensional features from the text content itself. Galende et al. (Galende et al., 2022) proposed the BORJIS deep learning framework, which achieves high-precision automatic identification of conspiracy theory content by employing a bidirectional GRU network on Twitter conversation-level text. However, such methods based on static word embeddings still have limitations when dealing with complex contexts. Transformer models represented by BERT and RoBERTa dynamically generate word embeddings based on context, significantly enhancing the model's ability to understand deep semantics (Patwardhan et al., 2023). Diab et al. (Diab et al., 2024)trained BERT-based models on Reddit corpora for cross-topic classification of conspiracy texts, with the RoBERTa model achieving optimal performance (0.714 F1 score and 0.787 AUC). Peskine et al. (Peskine et al., 2023) applied CT-BERT to the COVID-19 conspiracy theory detection task, achieving an F1 score of 0.810 and demonstrating the superior generalization capabilities of task-specific Transformer fine-tuning. Lei and Huang (Lei and Huang, 2023) built an event-aware language model based on Longformer and combined it with a heterogeneous graph attention network, achieving high-precision identification of conspiracy theory news in long documents. Existing research is gradually evolving from proxy detection based on shallow statistical features toward integrating deep semantic understanding.

3 Data

3.1 Dataset

This study employs the SemEval PsyCoMark dataset, which is sourced from more than 190 subreddits on the Reddit platform (Samory et al., 2025). The dataset consists of 4,316 labeled training instances, along with 100 development instances and 337 test instances, both of which are unlabeled. Regarding the data structure, each sample includes not only the core text, the conspiracy-theory classification label, and five categories of psycholinguistic markers (Actor, Action, Effect, Evidence, and Victim), but also retains key metadata including the sample ID, the originating subreddit, and the annotator information.

3.2 Data Preprocessing and Distribution Analysis

To ensure data quality, this study first cleaned the raw data: (1) **Deduplication:** Removed redundant samples with identical parameters (645 entries, 14.94%); (2) **Conflict Resolution:** Removed noisy samples with identical text but inconsistent conspiracy labels or markers (434 entries, 10.06%).

The distribution of the cleaned data is shown in Table 1. The 'Non-conspiracy' category is dominant (1,541 samples), followed by 'Conspiracy' (1,108) and 'Unknown' (588). Additionally, text-length analysis (see Appendix A1) reveals a distribution clustering between 30–60 words.

Further analysis highlights significant psycholinguistic distinctiveness: conspiracy texts show a 92% presence of 'Actor' and 'Action' markers, with substantially higher proportions of 'Effect', 'Evidence', and 'Victim' markers compared to non-conspiracy texts (e.g., 'Victim' is only 38% in the latter). This indicates that conspiracy theories rely on constructing complex causal chains with clear subjects and consequences.

4 Methodology

This study proposes a conspiracy analysis framework designed to accurately identify conspiracy content within unstructured social media text and further extract psychological markers.

4.1 Conspiracy Detection Module

To effectively capture the deep semantic context within social media posts and improve classification accuracy, we model conspiracy detection as a binary classification task (retaining only samples with explicit categories while filtering ambiguous instances). Building upon textual features, we construct a multimodel hybrid fusion architecture.

(1) Input Representation

Given a labeled dataset, the input comprises a text sequence T, a conspiracy label, and a set of metadata features M. The metadata set is defined as $M = \{m_{sub}, m_{src}, m_{ann}\}$, representing subreddit information, source ID, and annotator ID, respectively. The objective function aims to learn a mapping:

$$f(T, M) \to y \tag{1}$$

where $y \in \{0, 1\}$ represents non-conspiracy theory and conspiracy theory, respectively.

(2) Model Architecture

This study selected four representative BERT-based models to evaluate their performance from multiple dimensions (Qiu et al., 2020). BERT-Base (Devlin et al., 2019) serves as a baseline to assess improvements. DistilBERT (Sanh et al., 2019), incorporating knowledge distillation techniques, was chosen to investigate the efficacy of lightweight models. RoBERTa-Large (Liu et al., 2019) is optimized by eliminating the Next Sentence Prediction task; it aims to validate performance gains from increased capacity and optimized training. Finally, DeBERTa-V3 (He et al., 2021) integrates a Disentangled Attention mechanism, selected to validate improvements in feature extraction through relative position encoding.

Based on the selected models, this study established a processing workflow. The input text T is encoded by the BERT-based model to extract the special [CLS] token vector $h_{cls} \in R^d$ as the global semantic representation, while independent embedding layers are initialized for each discrete feature in the metadata set M. These multimodal features are integrated via concatenation:

$$\begin{aligned} h_{fusion} = \text{Concat}(&h_{cls}, E(m_{sub}), \\ &E(m_{src}), E(m_{ann})) \end{aligned} \tag{2}$$

Subsequently, the fused vector is normalized via a LayerNorm layer to mitigate gradient issues arising from differing feature distributions across modalities, and finally fed into a classification head consisting of a two-layer Multi-Layer Perceptron (MLP) with Dropout and ReLU activation functions to produce the predicted probabilities.

4.2 Conspiracy Marker Extraction Module

After detecting conspiracy texts, we further model the task as a token-level sequence labeling problem. This module aims to precisely locate and extract five key entity types: Action, Actor, Effect, Evidence, and Victim.

(1) Annotation Strategy

We adopt the BIO (Begin-Inside-Outside) annotation scheme, expanding the five entity types into 11 label categories (5 B-tags, 5 I-tags, and 1 O-tag). Special tokens in the model input (e.g., [CLS], [SEP], and padding tokens) are assigned specific ignore label IDs.

(2) Model Architecture

Adopting the four BERT-based models selected in the aforementioned conspiracy detection task,

Table 1: Statistics of conspiracy labels and markers.

Conspiracy Label	Total Texts	with Actor	with Action	with Effect	with Evidence	with Victim
Can't tell	588	424 (72%)	407 (69%)	288 (49%)	331 (56%)	252 (43%)
No	1,541	963 (62%)	955 (62%)	791 (51%)	726 (47%)	582 (38%)
Yes	1,108	1,015 (92%)	1,019 (92%)	857 (77%)	824 (74%)	755 (68%)
Total	3,237	2,402 (74%)	2,381 (74%)	1,936 (60%)	1,881 (58%)	1,589 (49%)

this module constructs an encoder-decoder architecture based on sequence labeling paradigm. The processing pipeline consists of three stages: First, Contextual Encoding employs the pre-trained models to extract the contextual hidden representation h_i for each token within the input sequence; Secondly, Token Classification involves stacking a fully connected layer (Linear Layer) on top of the Transformer output. This layer projects the hidden state of dimension $Hidden\_Size$ onto an 11-class label space. Finally, the decoding step applies the Softmax function to compute the probability distribution of each token, thereby precisely determining entity boundaries.

4.3 Experimental Setup

We adopt the model architectures detailed in Sections 4.1 and 4.2 as the foundation for this experiment. Building on this, we designed specific hyperparameters for both subtasks (details in Appendices B). Regarding the training workflow, given the unlabeled development and test sets, we employed a 9:1 hold-out split on the training data. This strategy enabled us to effectively monitor the training process and select the optimal epoch, thereby maximizing model performance. Furthermore, all experiments were conducted on the BwUniCluster 3.0 high-performance computing system using the gpu_a100_short partition.

4.4 Evaluation Metrics

The task organizer (Samory et al., 2025) established specific evaluation metrics for two subtasks. Subtask 1, Conspiracy Detection, is treated as a binary classification problem and uses Accuracy and Weighted F1-score as evaluation metrics. The Weighted F1-score calculates the weighted sum of precision and recall based on sample proportion, addressing potential class imbalance:

$$\text{Weighted F1} = \sum_{c \in \{Yes,No\}} w_c \times \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \quad (3)$$

For Subtask 2, Conspiracy Marker Extraction, employs a token-based Intersection over Union (IoU) strategy to accommodate boundary variations. A predicted segment is classified as a True Positive only when its entity type is correct and its token sequence achieves an IoU value of 0.5 or higher with the ground truth label. IoU is defined as follows:

$$\text{IoU}(S_{pred}, S_{gt}) = \frac{|T_{pred} \cap T_{gt}|}{|T_{pred} \cup T_{gt}|} \quad (4)$$

Based on this matching criterion, the Micro F1-score and Macro F1-score are calculated as the final metrics. The F1-score is the harmonic mean of Precision (P) and Recall (R). For a specific class i, it is defined as:

$$P_i = \frac{TP_i}{TP_i + FP_i}, \quad R_i = \frac{TP_i}{TP_i + FN_i} \\ F1_i = \frac{2 \cdot P_i \cdot R_i}{P_i + R_i} \quad (5)$$

The Macro F1 computes the unweighted arithmetic mean of F1-scores for each entity type, providing better insight into model performance on sparse or rare categories. It is defined as:

$$\text{Macro F1} = \frac{1}{n} \sum_{i=1}^{n} F1_i \quad (6)$$

The Micro F1 is derived from the global counts of TP, FP, and FN across all entity types, reflecting the model's overall extraction performance. It is defined as:

$$P_{micro} = \frac{\sum_{i=1}^{n} TP_i}{\sum_{i=1}^{n} TP_i + \sum_{i=1}^{n} FP_i} \\ R_{micro} = \frac{\sum_{i=1}^{n} TP_i}{\sum_{i=1}^{n} TP_i + \sum_{i=1}^{n} FN_i} \\ \text{Micro F1} = \frac{2 \times P_{micro} \times R_{micro}}{P_{micro} + R_{micro}} \quad (7)$$

5 Results and Analysis

5.1 Conspiracy Detection

Table 2 summarizes the performance of four BERT-based models on this task. In terms of overall per-

Table 2: Performance comparison of BERT-based models on the Conspiracy Detection task.

Model	Accuracy	Weighted F1	F1 Score (No)	F1 Score (Yes)
DistilBERT	**0.8052**	**0.8023**	**0.8544**	**0.7059**
BERT-Base	0.7922	0.7879	0.8462	0.6800
RoBERTa-Large	0.7662	0.7680	0.8163	0.6786
DeBERTa-V3	0.7662	0.7614	0.8269	0.6400

formance, DistilBERT achieved the best results, attaining the highest accuracy of 80.52% and a weighted F1 score of 0.80. The performance ranking of the models is as follows: DistilBERT > BERT-Base > RoBERTa-Large > DeBERTa-V3. This negative correlation between model size and performance suggests that larger models likely suffered from overfitting due to the limited size of the training dataset. All models demonstrated better performance on "non-conspiracy" (average F1 0.84) than on "conspiracy" (average F1 0.68). This gap indicates that accurately identifying positive samples (conspiracy) remains the primary bottleneck for current models.

5.2 Conspiracy Marker Extraction

For the subtask of Conspiracy Marker Extraction, Figure 2 illustrates the F1 score distribution across different models and entity types.

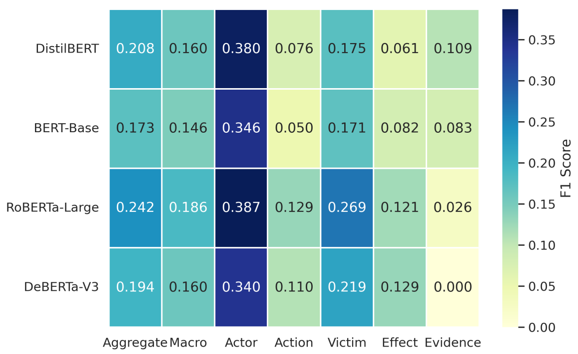

Figure 1: Performance Heatmap of Conspiracy Marker Extraction.

Overall, this task proved to be highly challenging, with limited performance observed across all models. Among them, RoBERTa-Large achieved the highest aggregated (micro) F1 score (0.242). However, all models performed at relatively low levels, showing the difficulty of accurately extracting conspiracy markers. Regarding specific marker types, the results reveal a distinct performance gap between concrete entities and abstract concepts. Models performed better on concrete entities (Actor and Victim) than abstract concepts (Action, Effect, and Evidence), with average F1 scores of approximately 0.38 and 0.20, respectively. However, model performance declined sharply when handling abstract concepts involving deep reasoning, indicating that current architectures struggle to capture such high-level semantic boundaries.

6 Conclusion

This study introduces the solution proposed for SemEval 2026 Task 10, including both conspiracy detection and marker extraction. In the Conspiracy Detection task, our model built upon DistilBERT demonstrated superior performance, achieving an F1 score of 0.80. By comparing BERT-based models with different architectures, we observed that lightweight models tend to achieve better generalization than large-scale models when data size is limited. In the Marker Extraction, models face challenges when handling abstract concepts involving deep logical reasoning. Experimental results indicate that existing architectures struggle to accurately capture the boundaries of high-level semantics.

Limitation Despite the improved performance in the detection task, the study remains limited by data quality and model characteristics. **Regarding data**, data scarcity and class distribution bias remain primary challenges. After removing duplicates and conflicting annotated samples, only over 3,000 effective training examples remain. Furthermore, the cleaned dataset exhibits significant category imbalance, with non-conspiracy categories accounting for 48% and conspiracy categories for 34%. **Regarding models**, BERT-based pre-trained models lack the capacity to comprehend deep semantics and abstract concepts when processing Reddit comments. Specifically, when extracting abstract conspiracy markers, models struggle to define clear semantic boundaries. Furthermore, lightweight models outperformed larger-parameter architectures in detection tasks. This suggests that under current limited data scales, large models may face overfitting challenges.

Future Work To address the aforementioned limitations in data and model inference, we will advance subsequent optimizations along two dimensions. **Regarding data**, we will implement data expansion and augmentation. First, we plan to integrate large-scale public conspiracy corpora to construct a more comprehensive domain dataset. Second, we will use Large Language Models (LLMs) for data augmentation by generating high-quality synthetic conspiracy samples or rewriting existing samples. **Regarding models**, we will innovate through adaptation and reconstruction. On the one hand, we will conduct domain-adaptive pre-training. Moving beyond the general BERT baseline, we will perform continuous pre-training based on collected specialized corpora. On the other hand, we will attempt to transform the task from simple token classification to prompt engineering based on LLMs, thereby achieving precise capture of the deep logical structure of conspiracy theories.

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A Distribution of Text Lengths

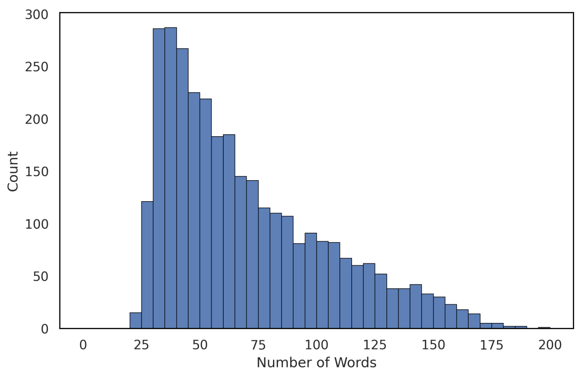

Figure 2: Distribution of text lengths (word count).

B Hyperparameter Configurations for Two Subtasks

Table 3: Hyperparameter configurations for two tasks.

Hyperparameter	Conspiracy Detection	Conspiracy Marker Extraction
Optimizer	AdamW	AdamW
Loss Function	Cross-Entropy	Cross-Entropy
Dropout	0.2 (Fusion), 0.1 (Hidden)	0.1
Learning Rate	2×10^{-5}	2×10^{-5}
Batch Size	16	32
Epochs	5	10
Max Sequence Length	256	256